

AKSHAY CHOUDHARY

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Education

National Institute of Technology, Rourkela

Bachelor of Technology in Biotechnology — CGPA: 8.59

Aug. 2023 – Present

Rourkela, Odisha

Relevant Coursework

Machine Learning | Statistics | Probability | Data Structures and Algorithms | Database Management Systems

Experience

Research Intern

May 2026 – Present

IIT Kharagpur (Grisha Research Internship Program)

Kharagpur, West Bengal

- Performed data preprocessing, feature representation, and statistical analysis for enzyme kinetics prediction datasets.
- Implemented, trained, and benchmarked deep learning models using PyTorch for enzyme kinetics prediction.
- Conducted experimental evaluation using R^2 , RMSE, and MAE metrics to assess model performance.
- Developed reproducible training, validation, and hyperparameter tuning pipelines for research workflows.

Projects

MiniML - Machine Learning Library | Python, NumPy, PyPI, GitHub, CI/CD

GitHub

- Developed and published a machine learning library implementing core supervised learning algorithms using NumPy.
- Designed Scikit-learn-inspired APIs for model training, evaluation, and inference.
- Implemented automated testing, package versioning, and CI/CD pipelines using GitHub Actions and Git workflows.
- Implemented machine learning algorithms including Linear Regression, KNN, K-Means, and DBSCAN using NumPy.

AI-Powered Meeting Minutes Generator | Pegasus, FastAPI, Hugging Face, React, Vercel

GitHub

- Built an end-to-end LLM-powered system for automatic meeting transcription and abstractive summarization.
- Developed data preprocessing pipelines for transcript cleaning, tokenization, and conversational normalization.
- Fine-tuned and evaluated Pegasus-based summarization workflows, improving summary quality measured through ROUGE evaluation.
- Designed and deployed FastAPI-based inference APIs for real-time transcript summarization and frontend integration.

Deep Learning for Enzyme Kinetics Prediction | PyTorch, Deep Learning, GNNs, Google Colab

GitHub

- Implemented the DLKcat architecture for enzyme turnover number (kcat) prediction using PyTorch.
- Integrated protein sequence embeddings and molecular graph representations for regression modeling.
- Reproduced baseline performance with R^2 of 0.46, RMSE of 3.67, and MAE of 2.72.
- Developed reproducible training and evaluation workflows for scientific machine learning experiments.

ML Fraud Detection System | XGBoost, Scikit-learn, Streamlit, Python

GitHub

- Built an end-to-end fraud detection pipeline using SQL, Pandas, and XGBoost on transactional datasets.
- Performed data cleaning, EDA, and feature engineering to improve model performance.
- Trained and compared Logistic Regression, Random Forest, and XGBoost models using ROC-AUC and F1-score.
- Developed an interactive Streamlit dashboard for real-time fraud risk monitoring and insights.

Technical Skills

Programming: Python, SQL, C++, C

Databases: MySQL, SQL Queries, Joins, subqueries, Window functions

Data Science: Pandas, NumPy, Data Analysis, Data Cleaning, EDA, Statistical Analysis, Matplotlib, Seaborn

Machine Learning: Scikit-learn, XGBoost, Random Forest, Classification, Regression, Feature Engineering, Model Evaluation, Cross Validation, Hyperparameter Tuning

Deep Learning: PyTorch

Development: FastAPI, Flask, REST APIs, React

Visualization: Matplotlib, Seaborn, Streamlit

Developer Tools: Git, GitHub, Linux, Docker, Jupyter Notebook

Open Source Contributions

Pylint Open Source Contributions | Python, Static Analysis, Open Source

2026

- Contributed production-grade fixes to the PyCQA/Pylint codebase.
- Merged pull requests into the PyCQA/Pylint codebase improving static analysis accuracy and Python compatibility.
- Collaborated with maintainers through code reviews and issue discussions.

Leadership / Extracurricular

- Organized HackOdisha 4.0 (100+ participating teams)
- Top 10 Finalist – Teach-a-thon Hackathon (500+ teams)
- Open Source Contributor – PyCQA/Pylint
- Solved 300+ algorithmic and data structure problems across competitive programming platforms.